
Homework 2: VAE for Chest X-ray Image Generation

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1 Hyperparameters Setting

Table 1 summarizes the main training hyperparameters.

Table 1: Training hyperparameters.

Parameter	Value
Image size	64×64 (grayscale)
Latent dimension	128
Batch size	64
Optimizer	Adam
Learning rate	2×10^{-4}
Epochs	50
β (KL light)	1.0

2 Results

2.1 Training Loss Curves

Loss curves are shown in Figure 1. The total loss decreases rapidly in early epochs and gradually reaches a plateau. The reconstruction term dominates due to pixel-wise summation, while the KL term stays comparatively small.

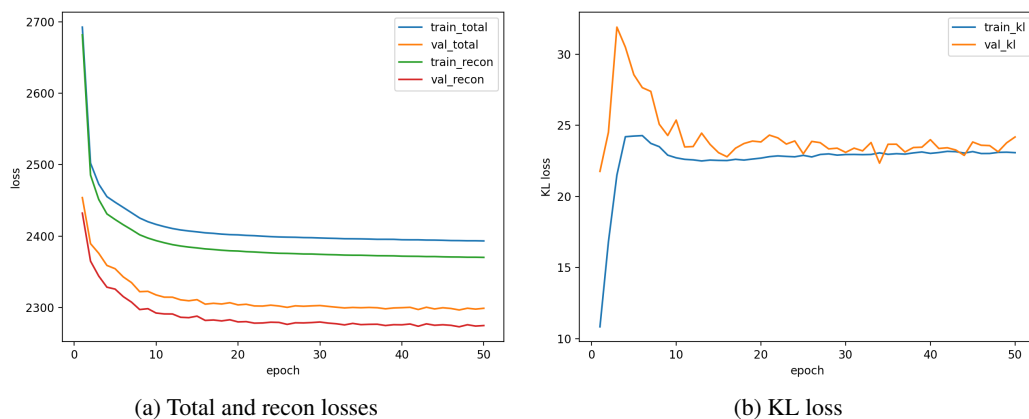


Figure 1: Training/validation loss curves. (a): total and reconstruction. (b): KL term.

2.2 FID and Inception Score

I initially followed the reference implementation linked in the homework (PyTorch-Ignite blog) to compute FID/IS, but obtained an unexpectedly small FID (0.1348), which is inconsistent with the

visibly blurry VAE generations. This behavior is consistent with prior reports[1] that Ignite’s FID metric can differ from reference implementations by several orders of magnitude. As a result, I re-implemented the same Inception-v3 feature-based evaluation pipeline in a standalone script (i.e., extracting Inception features for real and generated samples and computing the Fréchet distance), and report the results below. The relatively large FID further reflects the gap between generated and real images.

Table 2: FID and Inception Score on generated samples.

Metric	Value
FID ↓	338.5890
Inception Score ↑	1.8136 ± 0.0575

2.3 Qualitative Results: Real vs. Reconstructions vs. Samples

Figure 2 compares real validation images, reconstructions (encoding and decoding the same inputs), and unconditional samples decoded from $z \sim \mathcal{N}(0, I)$. Reconstructions preserve global structure but appear blurry, which is common for VAEs trained with pixel-wise likelihood objectives.

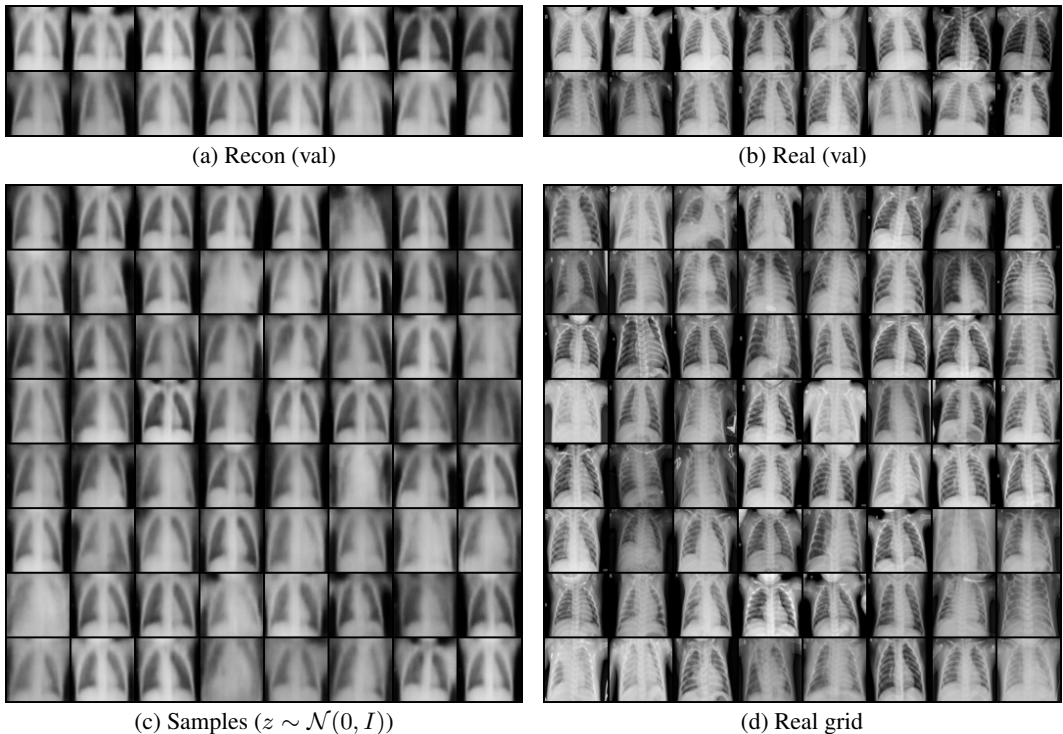


Figure 2: Qualitative comparison at epoch 50. (a) Reconstructions of real images. (b) Real validation images. (c) Random samples generated by decoding $z \sim \mathcal{N}(0, I)$. (d) A larger grid of real images for reference.

These results above show that VAE converges stably and produces samples with plausible coarse chest structure, but outputs are noticeably blurry. This is consistent with the tendency of VAEs trained with pixel-wise reconstruction losses to average over high-frequency details. Personally speaking, potential improvements might be higher resolution training, Gaussian or MSE likelihood for continuous intensities, KL warm-up, or stronger decoders to better capture fine anatomical details.

References

- [1] pytorch/ignite contributors. Fid metric returns different value than pytorch-fid. <https://github.com/pytorch/ignite/issues/2423>, 2021. GitHub issue #2423, accessed 2026-02-11.